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ASSIGNMENT 1

REPORT WRITING

SPECIAL TOPICS IN DATA ENGINEERING

LECTURER: DR ARYATI BAKRI

GROUP 18

MEMBERS:

NAME	MATRIC. NO
SYARIFAH DANIA BINTI SYED ABU BAKAR	A23CS0183
YASMIN BATRISYIA BINTI ZAHIRUDDIN	A23CS0201

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DATA WAREHOUSE, DATA LAKE AND DATA LAKEHOUSE

Syarifah Dania Binti Syed Abu Bakar , Yasmin Batrisyia Binti Zahiruddin[2]

[1]Faculty of Computing & University of Technology Malaysia (UTM), Malaysia

[2]Faculty of Computing & University of Technology Malaysia (UTM), Malaysia

[1]syarifahdania@graduate.utm.my ; [2]yasmin.batrisyia@graduate.utm.my

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1.0 Executive Summary

Data lakes, data warehouses and data lakehouses are three essential data management technologies that are examined in this report. Organizations use these systems to process, store and analyze vast amounts of data. The report discusses practical applications, contrasts the benefits and drawbacks of each technology and suggestions are offered to assist organizations in selecting the best option for their requirements.

2.0 Introduction

Big data analytics is one of the most famous words in today's digital world. This is because examining big data helps to find and examine the correlation, and hidden pattern that is available in the data. There are several types of data repositories in the enterprise that stores the data which are Data Lake, Data Warehouse, and Data Lakehouse.

Referring to Nambiar and Mundra (2022), Data Lake is a centralized place that receives raw data from any format (structured, semi-structured or unstructured) without any standardised schema or purpose. Meanwhile, Data Warehouse is a purpose-built repository that stores data in a structured, filtered and pre-processed data organized under standardised schemas for specific analysis purposes. Furthermore, Data Lakehouse stores both unstructured and structured data processing from the same dataset (Rodríguez-Mazahua et al., 2026).

This data repository is needed and relevant to the enterprise system. For example, a data warehouse could give a stable and curated store of structured data so that the data is reliable to make a report, business-intelligence and decision-making. Moreover, by using data repositories such as data lake, it allows enterprises to store the full use information that are generated by IoT devices, social media, logs and legacy systems. Moreover, said Rodríguez-Mazahua et al. (2026), Data lakehouse is relevant because it could offer scalability, cost efficiency, standardised analytics and cloud-native integration. Therefore, this paper paper aims to analyze deeper about data lake, data warehouse, and data lakehouse as these technology keeps on evolving nowadays in the enterprise system by explaining how these technology is actually works, providing the example case study, listed advantages and disadvantages, stated the challenges occur from these technologies, and provide solutions to it.

3.0 Literature Review

The rapid growth of digital technologies has led to an exponential increase in data generation because it comes in vast amounts, at high speed and in many different forms (BENTAIB et al., 2024). To deal with these problems, creation of data lakes, data warehouses and data lakehouses.

3.1 Data Lakes

The conception of data lakes is still quite vague in both academic and industry conversations. The term 'data lake' was first used by James Dixon in 2010 to define a system that can store and hold a massive amount of raw data in its original format such as structured, semi-structured and unstructured data for later analysis (BENTAIB et al., 2024).

The ability of data lakes to store raw data in many formats provides a greater flexibility for big machine learning projects and preprocessing data in real time (Reddy Rella, 2025). Data lakes use a schema-on-read approach which means that data is only organized when it is accessed, not when it is stored. Researchers say that data lakes are scalable systems that use metadata and unique identifiers to organize data.

Data lakes lets businesses do advanced analytics, machine learning and preprocessing in real time due to its flexible and scalable nature. Data lakes provide better data collection, faster access to raw data, hence less work is needed to process data at first.

3.2 Data Warehouses

According to SAP, n.d, data warehouse can be defined as a centralized digital storage system that is made to gather, combine and handle vast amounts of data from many sources. They are made to hold processed, cleaned and standardized or converted structured data for use in reporting and business intelligence (BI). Data warehouses apply a preset schema to combine data from many sources, guaranteeing consistency and dependability for analysis (Kosinski, 2025).

Data warehouses mainly concentrate on storing historical, summarized, and consolidated data. These systems create remarkably huge storage systems, frequently reaching from hundreds of terabytes to petabytes, by integrating data from several operational databases over lengthy periods of time. Data warehouses are built to facilitate sophisticated analytical queries rather than standard transaction processing as a result of this consolidation (Kosinski, 2025).

3.3 Data Lakehouses

The data lakehouse, a contemporary data architecture, combines the advantages of data lakes and data warehouses. It combines the performance capabilities and organized data management of data warehouses with the adaptability and scalability of data lakes. A data lakehouse offers a unified platform that supports many data formats and permits analytics, reporting, and machine learning inside a single system, according to Amazon Web Services (2024). Data lakehouses also solve important problems with conventional systems, including data duplication, system complexity and delays brought on by transferring data between platforms. Lakehouses increases productivity and facilitate features like ACID transactions, metadata management and Medallion data architecture that consists of bronze (raw), silver (filtered) and gold (curated) data as shown in Figure 2 for improved organization and governance by combining data into a single system (Armbrust et al., 2021).

4.0 Analysis and Discussion

This chapter will elaborate in detail about how these technologies work, the case example, advantages and disadvantages and the challenges that occur when implementing it.

4.1 How the Technology Works

These data repositories work differently. Based on Harby and Zulkernine (2024), it says that Data warehouse apply a classic ETL pipeline where they extract the data from operation sources, transform to a predefined relational schema, and load it into fact and dimension table structures for Business Intelligence (BI) reporting. The type of schema is a schema on write, so that every row following the model, applied strong ACID guarantees and fast SQL based analytics. For example, queries that typically execute are by HiveSQL that will later translate them into MapReduce or Spark jobs for petabyte scale batch analysis.

Next, they also mentioned that data lake extract raw data in its initial form where it can be structured, semi-structured or unstructured data by using ELT or streaming tools, and put it in a distributed file system such as HDFS without changing the schema. Data lake applied a schema on read where it means the structure can only be interpreted when a query accesses the files. This support batch, near real time and real time processing.

Lastly, they explained Data Lakehouse combines the scalable storage of a lake with the governance and ACID properties of a warehouse. Data is stored as open format files along with transaction logs that record all activities because the log could provide reliable metadata. With that, the system can enforce schema at read-time while still supporting batch data, streaming workloads, and delivering fast OLAP query performance.

In conclusion, these technologies work differently with different executions. Data Warehouse focuses on curated, structured data for analytics, Data Lake focus on raw, and flexible ingestion, meanwhile Data Lakehouse supports both without moving data between separate systems.

4.2 Case examples

Data Warehouse, Data Lake, and Data Lakehouse have been applied to many sectors to help make better BI reporting. The example of a Data Warehouse application is the Covid Warehouse where it implements the classic data warehouse pipeline. It turns the daily Italian COVID-19 CSV files into a unified, query ready store. The covid data warehouse functioned by went through integration level where it runs the automatic ETL in Python (Pandas). Secondly, Warehouse level stores the cleaned table. Next, is the analysis level where it automatically creates multidimensional data cubes. Lastly , the visualization layer is where generating a visual graphic to display light strong positive or negative relationships (Agapito et al., 2020).

The next example is the data lake that was applied in QueryArtisan. It treats the data lake as a unified storage hub that includes heterogeneous datasets. It not only stores the raw data but it also gives the meta information and processing resources for analytics and ETL pipelines. (Liu et al., 2025).

Lastly, Data Lakehouse is applied to solve modern Big Data challenges by building a proof of concept implementation of a modern data platform using DataOps methodologies for process kubernetes as an orchestration platform and data application by using Cloud-Native software. (AbouZaid et al., 2025)

4.3 Advantages and Disadvantages

This section discusses the key advantages and disadvantages of data warehouses, data lakes, and data lakehouses.

4.3.1 Data Lake

Data lakes provide a very adaptable and scalable data storage. As a result, they are very helpful for machine learning, artificial intelligence and big data analytics applications, they are also affordable because they employ inexpensive storage techniques. However, in the absence of adequate governance, data lakes may turn into chaotic 'data swamps' causing low-quality data and challenging retrieval and ingestion. Furthermore, they are not designed for quick querying and frequently need technical expertise to be used effectively (Nambiar & Mundra, 2022).

4.3.2 Data Warehouses

Data warehouses offer robust performance and dependability for structured data analysis. They are appropriate for BI, reporting and decision-making since they are geared for complicated queries. It also guarantees consistency, correctness and robust governance since data is cleaned and organized before being stored. However, their implementation and maintenance can be expensive, particularly when working with large-scale data sets. Additionally, when managing unstructured or rapidly changing data, their inflexible schema design reduces flexibility (Kosinski, 2025; AWS, 2024).

4.3.3 Data Lakehouses

Data lakehouses increase productivity and eliminate data duplication by carrying out the needs for separate systems through structured reporting and advanced analytics in a single environment. Reliability and governance are enhanced by features like layered architectures, metadata management and ACID transactions. Yet, some businesses find it difficult because they are an emerging architecture that can be difficult to install and require specialized knowledge and expertise (Rodríguez-Mazahua et al., 2026)

4.4 Challenges

Implementing data lakes, data warehouses, and data lakehouses brings a number of challenges for organizations. Data governance and data quality management are two significant challenges. Specifically, if appropriate governance is not implemented, data lakes are likely to become chaotic, resulting in inconsistent, redundant, or unreliable datasets. Maintaining data quality and consistency across various sources is still necessary, even in more organized systems (Kosinski, 2025; AWS, 2024)

System integration and complexity present another major difficulty. The technical complexity of modern data architectures is typically increased by the need to combine many tools, storage systems, and processing frameworks. Redesigning current pipelines and modifying organizational workflows may also be necessary when switching from conventional data warehouses or data lakes to a unified lakehouse architecture. Higher installation effort and operational costs could occur from this (Armbrust et al., 2021).

Last but not least is the requirement for qualified technicians and resources. To properly manage infrastructure and guarantee peak performance, these systems require knowledge of data engineering, cloud computing, and analytics. Organizations may find it difficult to fully exploit complex functions like centralized processing and metadata management if they lack the necessary technical expertise.

5.0 Recommendations

With the challenges and disadvantages that has been mentioned above, the recommendation of solutions to propose are by Harby and Zulkernine (2024), where it says to implement Data Lakehouse architecture such as Amazon S3 because it can also provide data storage solutions with high accessibility, scalability, and cost effectiveness.

Next the recommendation made for future direction for Data Warehouse is by applying a cloud data warehouse. It can help to overcome the huge cost of purchasing, infrastructure, installation, etc. The long term gains of cloud data warehouse are mainly data availability and scalability. The flexibility of cloud based service enables a very wide distribution of cloud services. To improve data management and discovery on a large scale, computer intelligence and big data have to work together with IoT as well (Nambiar & Mundra, 2022).

Lastly, the recommendations proposed by Nambiar and Mundra (2022) for data lake are by focusing on Machine Learning application toward data set organization and discovery. The discovery task usually involves finding similar attributes from the data, metadata and further being done classification or clustering tasks. For example, techniques that can be used are KNN classifier and a logistic regression model for optimizing feature coefficients.

6.0 Conclusion

In conclusion, data warehouses, data lakes and data lakehouses are important data management frameworks that facilitate organizational decision-making and contemporary big data analytics. While data lakes offer flexible and scalable storage for raw structured, semi-structured, and unstructured data, data warehouses are intended for structured, high-quality data and effective business intelligence reporting. By combining the two methods, the data lakehouse allows for unified analytics, machine learning and storage on a single platform. These systems enhance data processing and decision-making, but they also come with drawbacks like complicated systems, expensive installation costs, and governance concerns. Therefore, while choosing an appropriate architecture, firms must carefully consider their data needs, technical capabilities and long-term objectives. The ongoing development of these technologies emphasizes how crucial effective, scalable and integrated data solutions are to assisting data-driven businesses in the current digital era.

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APPENDIX

Figure 1 shows Data Warehouse, Data Lake, and Data Lakehouse Architecture.

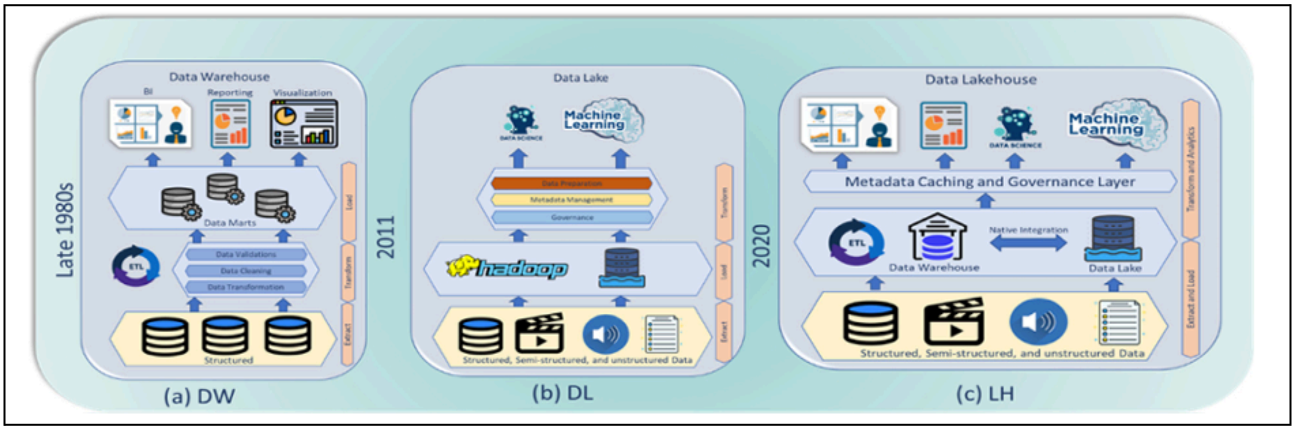

Figure 1

Figure 2 shows a Medallion Data Architecture.

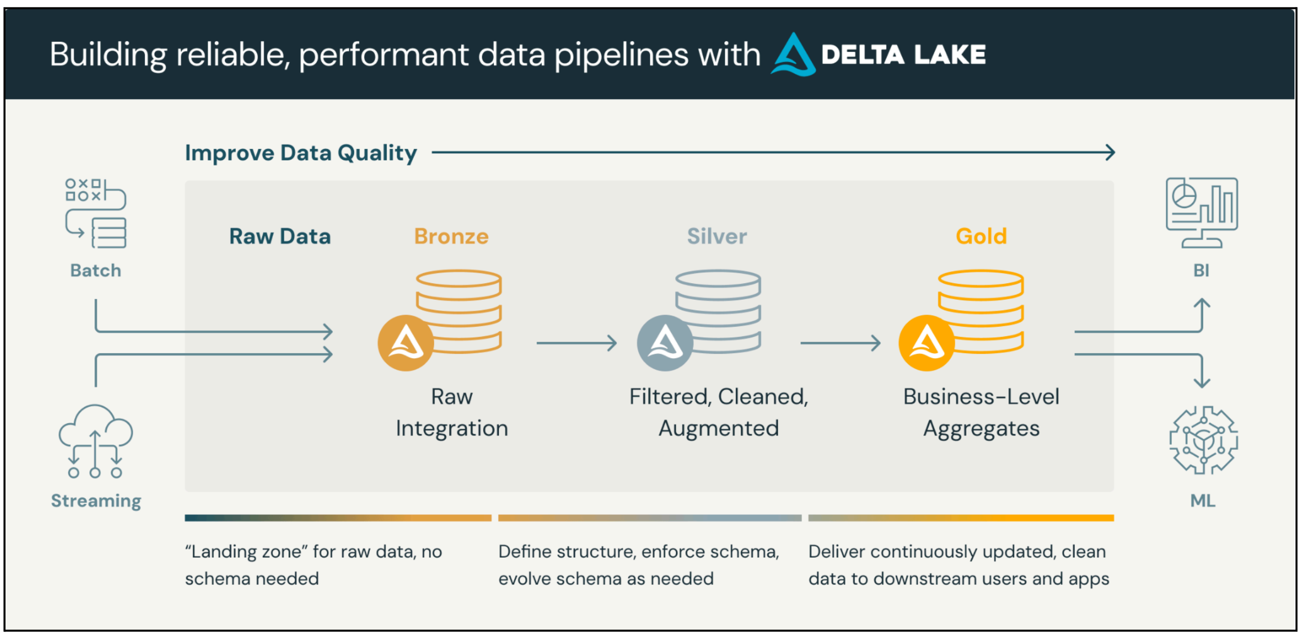

Figure 2